Calculus II

Training Problems #1 17 – 21 Feb 2025

◇ **Topics:**

- Functions of several variables
- Level curve, level surface
- Limit

In Problems 1–5, find the limits of functions of two variables.

Problem 1. $\lim\limits_{(x,y)\to(0,0)} \dfrac{3x^2 - y^2 + 5}{x^2 + y^2 + 2}$.

Problem 2. $\lim\limits_{(x,y)\to(3,4)} \sqrt{x^2 + y^2 - 1}$.

Problem 3. $\lim\limits_{(x,y)\to(0,\ln 2)} e^{x-y}$.

Problem 4. $\lim\limits_{(x,y)\to(0,0)} \dfrac{e^y \sin x}{x}$.

Problem 5. $\lim\limits_{(x,y)\to(1,\pi/6)} \dfrac{x \sin y}{x^2 + 1}$.

In Problems 6–8, find the limits by rewriting the fractions first.

Problem 6. $\lim\limits_{(x,y)\to(1,1)} \dfrac{x^2 - 2xy + y^2}{x - y}$.

Problem 7. $\lim\limits_{(x,y)\to(1,1)} \dfrac{xy - y - 2x + 2}{x - 1}$.

Problem 8. $\lim\limits_{\substack{(x,y)\to(2,0)\\ 2x-y\neq 4}} \dfrac{\sqrt{2x - y} - 2}{2x - y - 4}$.

In Problems 9–10, find the limits of functions of three variables.

Problem 9. $\lim\limits_{P\to(1,3,4)} \left(\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z}\right)$.

Problem 10. $\lim\limits_{P\to(\pi,0,3)} ze^{-2y} \cos 2x$.

END

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Answer to selected problems

1. $\frac{5}{2}$
2. $2\sqrt{6}$
3. $\frac{1}{2}$
4. 1
5. $\frac{1}{4}$
6. 0
7. -1
8. $\frac{1}{4}$
9. $\frac{19}{12}$
10. 3